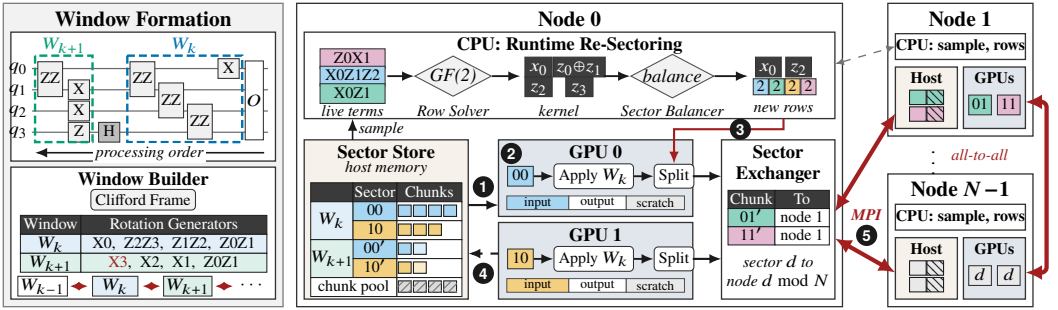

Window Formation
W_{k+1}
W_k
q_0
q_1
q_2
q_3
ZZ
X
Z
H
O
processing order
Window Builder
Clifford Frame
Window
Rotation Generators
W_k
X0, Z2Z3, Z1Z2, Z0Z1
W_{k+1}
X3, X2, X1, Z0Z1
W_{k-1}
W_k
W_{k+1}
Node 0
CPU: Runtime Re-Sectoring
Z0X1
X0Z1Z2
X0Z1
live terms
GF(2)
Row Solver
x_0 $z_0 \oplus z_1$
z_2 z_3
kernel
balance
Sector Balancer
x_0 z_2
2 2 2 2
new rows
sample
Sector Store
host memory
Sector
Chunks
W_k
00
10
W_{k+1}
00′
10′
chunk pool
GPU 0
00
Apply W_k
Split
input
output
scratch
GPU 1
10
Apply W_k
Split
input
output
scratch
Sector Exchanger
Chunk
To
01′
node 1
11′
node 1
sector d to
node d mod N
MPI
all-to-all
Node 1
CPU: sample, rows
Host
GPUs
01
11
Node N −1
CPU: sample, rows
Host
GPUs
d
d